

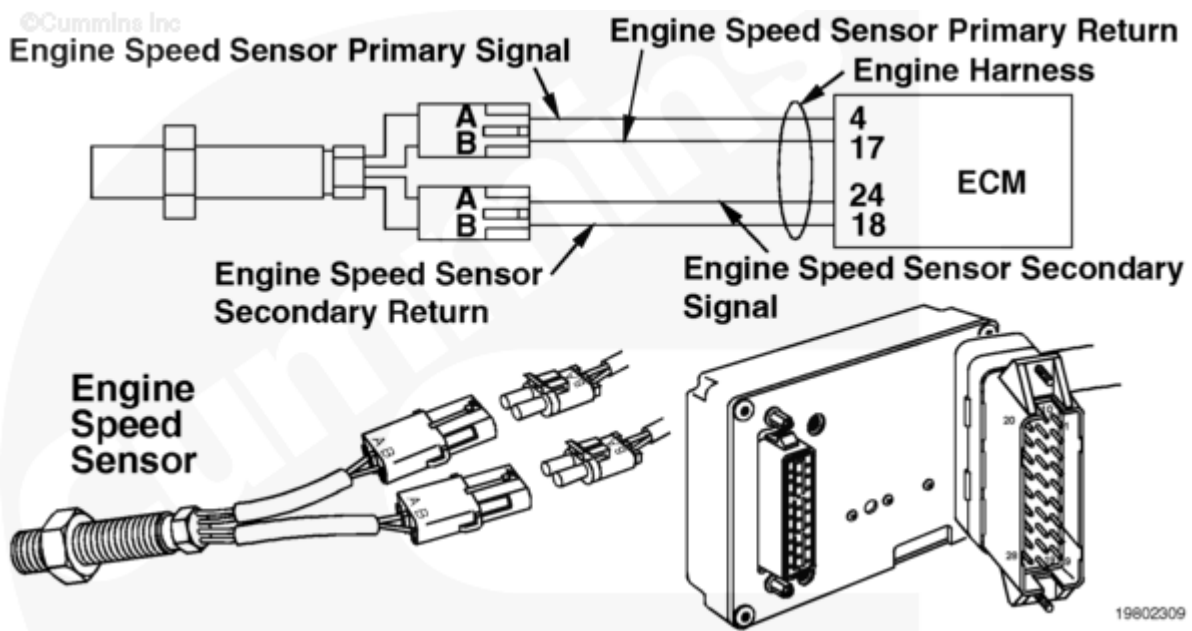
Fault Code: 121

Engine Magnetic Crankshaft Speed/Position Lost One of Two Signals - Abnormal Rate of Change

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 121 PID(P): 190 SPN: FMI: 10 Lamp: Flashing SRT: 00-621	No engine speed signal detected from either the main or backup speed sensors.	None on performance.



Engine Speed Sensor Circuit

Circuit Description

The engine speed sensor is dual channel speed sensor used by the ECM to monitor the engine speed. The speed sensor has four circuits: two signal circuits, and two return circuits. As the teeth on the crankshaft flywheel move past the speed sensor, a signal is generated on the speed sensor signal circuits.

Component Location

The engine speed sensor is located on the flywheel housing.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected a signal failure with engine speed sensor.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostic runs and fails.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- Engine will run on backup speed signal provided by the camshaft position sensor.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.

- Malfunctioning or damaged engine speed sensor.

Refer to Troubleshooting Fault Code t05-121 (</qs3/pubsys2/xml/en/procedures/98/98-t05-121.html>)

Last Modified: 08-Sep-2021
